Six Birds: No-Go Theorems for Audited Emergence

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Abstract

When a system is viewed at a coarser scale, it is tempting to say that something new has emerged: a direction of time, a driving force, an autonomous macroscopic law, a stable object, or an open-ended hierarchy of levels. Such claims are easy to make and hard to check. We study them in the simplest setting in which they can be checked exactly: finite Markov chains observed through deterministic maps, or *lenses*. In this setting we prove eight no-go theorems. Coarse observation can never increase the arrow of time, measured as the relative entropy between the path law and its time reversal. A chain started in a law that satisfies detailed balance shows zero arrow under every lens, however rich the hidden protocol structure it carries. On a graph without cycles every antisymmetric edge field is the gradient of a potential, and gradient fields sum to zero around every closed walk, so a force-like cycle affinity needs both a cycle and a non-gradient field. For a lumped chain, the best macroscopic transition kernel at a given lag is the conditional law of the future macrostate, and its unavoidable loss is a conditional mutual information, the *closure deficit*; it is strictly positive as soon as two microstates of positive mass in the same macrostate have different macroscopic futures. Under strict Dobrushin contraction with coefficient $\lambda < 1$, any two ε-approximately stationary laws lie within $2\varepsilon/(1-\lambda)$ of each other in total variation, so such dynamics has at most one stationary law and cannot carry two approximately stationary laws farther apart than this bound. Finally, idempotent packaging stabilizes after a single step, and a fixed interface with finite image defines exactly $2^{|\mathrm{im}(f)|}$ predicates, so no unbounded ladder of new distinctions can be built inside it. All eight theorems are formalized and machine-checked in Lean 4 with Mathlib.

Keywords: coarse-graining; finite Markov chains; entropy production; data-processing inequality; cycle affinity; lumpability; closure deficit; Dobrushin contraction; formal verification.

1 Introduction

Many claims about emergence share one pattern. A system is specified at a fine, microscopic level. It is then viewed through a coarse description, and the coarse description seems to show something the microscopic specification never mentioned: a direction of time, a force that drives currents around loops, a macroscopic law that runs on its own, a persistent object, or a hierarchy of levels that keeps producing new distinctions. Whether such a claim is true depends on details that informal arguments tend to blur. Which coarse description is meant? Which quantity is computed from it? Is that quantity computed from the model itself, or from a convenient approximation fitted to it?

This paper answers these questions in the simplest setting where every answer can be checked exactly. The microscopic system is a finite Markov chain, the coarse description is a deterministic map from microstates to macrostates (a *lens*), and every quantity is an exact finite expression in

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the transition probabilities, the initial law, and the lens. We call such quantities *audits*. Within this setting we prove eight theorems, each of which rules out one way of making something appear that is not already present. The theorems are elementary, but the hypotheses are exact, and the hypotheses are what matter: each theorem says what an emergence claim must add before it can succeed.

Exact observation, not fitted proxies. Throughout, a coarse quantity is computed from the *honest* observed process $Y_t = f(X_t)$, whose path law is the pushforward of the microscopic path law along the lens. A common shortcut replaces the observed process by a first-order Markov chain fitted to it on the macrostates. The fitted chain is a different object. It can be a useful diagnostic, but none of the theorems below is about it, and conclusions drawn from it are not covered by them.

The eight results. Table 1 summarizes the theorems; precise statements are collected in Section 3. They fall into five groups.

1. *Arrow of time* (Section 4). The arrow of a process is the Kullback–Leibler divergence between its path law and the law of the time-reversed paths, a path-space measure of irreversibility; for a stationary chain it is the entropy production over the observation window [14, 17]. Observation through a lens can only decrease it (Theorem 3.1). If the chain starts in a law satisfying detailed balance, the microscopic arrow is zero, so every observed arrow is zero as well, however much protocol-like structure hidden coordinates impose on the observed process (Theorem 3.2).

2. *Forces and cycles* (Section 5). On a graph without cycles, every antisymmetric edge field is the gradient of a potential (Theorem 3.3), and gradient fields sum to zero around every closed walk (Theorem 3.4). Applied to the log-ratios $\log\big(P(x,x')/P(x',x)\big)$ of a Markov kernel, this says that a nonzero cycle affinity [16] requires a cycle in the support and a log-ratio field that is not a gradient.

3. *Closure of macroscopic laws* (Section 6). For a given current law, lens, and lag, the best macroscopic kernel is the conditional law of the future macrostate given the present macrostate, and the loss it cannot avoid is a conditional mutual information, the closure deficit (Theorem 3.5). The deficit is strictly positive as soon as two microstates of positive mass in the same macrostate have different macroscopic futures.

4. *Objects under contraction* (Section 7). If the dynamics is a strict contraction in total variation, any two approximately stationary laws are close, with an explicit bound, and there is at most one stationary law (Theorem 3.6). A contractive dynamics therefore cannot carry two robust objects separated by more than that bound.

5. *Ladders of levels* (Section 8). Applying an idempotent packaging map more than once gains nothing (Theorem 3.7), and a fixed interface with finite image supports exactly $2^{|\mathrm{im}(f)|}$ definable predicates, so it cannot support an unending sequence of new distinctions (Theorem 3.8).

A vocabulary of primitives. The theorems belong to the Six Birds program [21], which describes how new levels of description arise in terms of six primitive operations that can be added to a finite model: *rewrite* (updates that change which microstate is represented), *gating* (deleting or suppressing transitions), *holonomy or protocol* (hidden cyclic or scheduling coordinates that the observation may not see), *staging or refinement* (splitting states to add internal degrees of freedom), *packaging* (maps that compress states or distributions, such as idempotent endomaps and contractive kernels), and *drive* (explicit breaking of detailed balance). The primitives are used here only as a vocabulary for

Table 1. The eight theorems at a glance. The last column names an emergent feature that the theorem constrains and a condition the feature requires. In the null-affinity and closure rows the condition is also sufficient; in the other rows it is necessary but not sufficient.

Theorem	Conclusion	Feature and what it requires
3.1 Arrow monotonicity	Observed arrow $\leq$ microscopic arrow, for every lens, initial law and horizon.	A positive observed arrow requires a positive microscopic arrow.
3.2 Protocol trap	A chain started in a detailed-balance law has zero observed arrow under every lens.	A positive observed arrow requires an initial law not in detailed balance with the kernel.
3.3 Forest exactness	On a forest, every antisymmetric edge field is a potential gradient.	A nonzero cycle affinity requires a cycle in the support graph.
3.4 Null cycle affinity	A gradient field sums to zero around every closed walk.	A nonzero cycle affinity occurs exactly when the log-ratio field is not a gradient.
3.5 Closure deficit	Best macro kernel is the conditional macro law; minimal loss equals $I(X_0; Y_\tau \mid Y_0)$.	A positive deficit occurs exactly when positive-mass microstates in one macrostate have different macro futures.
3.6 Contractive objecthood	Two ε-stable laws are within $2\varepsilon/(1-\lambda)$; at most one stationary law.	Two distinct stationary laws require $\lambda(P) = 1$.
3.7 Idempotent saturation	$e \circ e = e$ implies $e^{\circ n} = e$ for all $n \geq 1$.	An unending ladder of iterates requires a packaging map that is not idempotent.
3.8 Bounded interface	Exactly $2^{\lvert\mathrm{im}(f)\rvert}$ definable predicates; no infinite sequence of distinct ones.	An unending sequence of new distinctions requires leaving a single fixed interface of finite image.

contrasts; they are neither mutually exclusive nor exhaustive. In this vocabulary, several theorems say that one primitive cannot stand in for another. Hidden protocol coordinates do not produce an arrow of time without a drive or a nonequilibrium start (Theorem 3.2). A cycle affinity needs both cycles in the support and a drive that makes the log-ratio field non-exact (Theorem 3.3, Theorem 3.4). Packaging alone does not generate an unbounded hierarchy: idempotent packaging saturates at once, and a fixed interface saturates after finitely many distinctions (Theorem 3.7, Theorem 3.8).

Formal verification. All eight theorems are formalized in Lean 4 with Mathlib. The formal statements use real-valued probability laws, extended-real Kullback–Leibler divergence (so infinite divergences are covered rather than excluded), genuine Markov path laws, and Mathlib's simple graphs. The proofs are kernel-checked and depend only on Lean's standard axioms. Appendix E matches each theorem with its formal counterpart and records where a formal statement is more general than the version in the text, or where a supporting argument in the text is not part of the formal development.

Relation to earlier work. The individual facts behind several theorems are classical: the data-processing inequality for relative entropy [3], the time-reversal symmetry of reversible chains [9], the cycle-affinity picture of Markov networks [16], lumpability of Markov chains [2, 10], and Dobrushin's contraction coefficient [6, 18]. The contribution here is to state them for one fixed set of exact audits, with hypotheses sharp enough to separate the theorem object from its common proxies, and to verify them formally. Earlier papers in the Six Birds series developed the semantics of packaging and coarse-graining [22], protocol traps [23], closure deficits [19], agenthood [24], and the classification of layer births [20]; they serve as background, and the present paper does not depend on them.

Organization. Section 2 fixes notation and defines the audits. Section 3 states the eight theorems. Sections 4 to 8 prove them, give examples, and discuss their scope. Section 9 draws the threads together. Routine calculations are in Appendices A to D, the formalization map is in Appendix E, and Appendix F records the version history. A separate supplement documents provenance and the computational audit artifacts; the statements and proofs in this paper do not depend on it.

2 Setting and audits

All sets in this section are finite unless stated otherwise. Figure 1 gives a picture of the setting.

2.1 Laws, kernels, and path laws

For a finite set X, write $\Delta(X)$ for the set of probability laws on X, and $\mu(A) = \sum_{x \in A} \mu(x)$ for $A \subseteq X$. A *Markov kernel* on X is a matrix $P : X \times X \to [0, 1]$ whose rows sum to one. A law μ is moved one step forward by $(\mu P)(x') = \sum_x \mu(x) P(x, x')$. A law π is *stationary* if $\pi P = \pi$; every kernel on a nonempty finite set has at least one stationary law. A law π satisfies *detailed balance* with P if

$$\pi(x) P(x, x') = \pi(x') P(x', x) \qquad \text{for all } x, x' \in X.$$

Summing over x shows that such a π is stationary. These are the standard notions of finite Markov chain theory [12, 15].

Fix an initial law μ, a kernel P, and a horizon $T \geq 0$. The *path law* on X^{T+1} is

$$\mathbf{P}_T^{\mu,P}(x_0, \ldots, x_T) = \mu(x_0) \prod_{t=0}^{T-1} P(x_t, x_{t+1}),$$

the law of the trajectory $(X_0, \ldots, X_T)$ of the chain started from $X_0 \sim \mu$. Path reversal $\mathrm{rev}(x_0, \ldots, x_T) = (x_T, \ldots, x_0)$ is an involution of X^{T+1}, and the *reversed path law* is its pushforward,

$$\mathrm{rev}_{\#} \mathbf{P}_T^{\mu,P}(\gamma) = \mathbf{P}_T^{\mu,P}(\mathrm{rev}\gamma).$$

It gives each path the probability that the forward process assigns to the same path traversed backwards. For a stationary reversible chain this is the classical time-reversal construction [9]; we use it for every initial law.

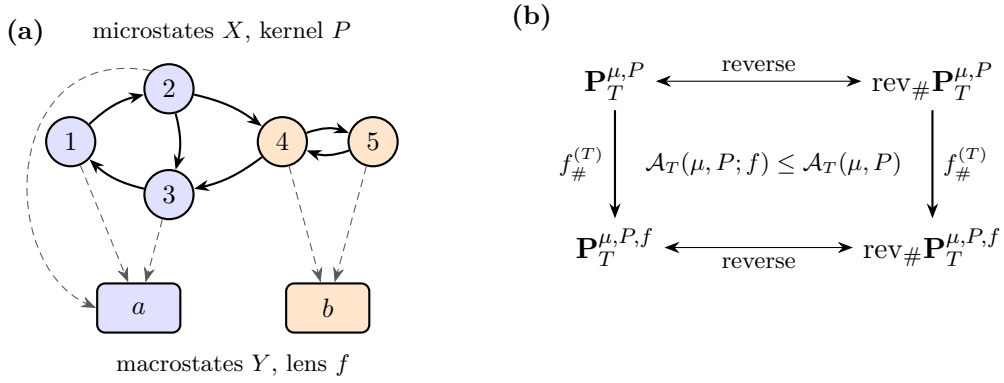


Figure 1. The setting. (a) A finite Markov chain on microstates is observed through a lens f that merges microstates into macrostates; the observed process is $f(X_t)$. (b) Reversing paths and applying the lens commute (Lemma 4.2). The arrow is the divergence between the top row of laws, and the observed arrow the divergence between the bottom row, so the data-processing inequality yields Theorem 3.1.

2.2 Lenses and observed processes

A *lens* is a function $f : X \to Y$; other papers in the series write a lens as ℓ, or as Π when it packages microstates into macrostates.[1] The *pushforward* of $\mu \in \Delta(X)$ is $(f_{\#}\mu)(y) = \mu(f^{-1}(y))$, and the sets $f^{-1}(y)$ are the *fibers* of f. Applying f at every time gives a map on paths, $f^{(T)}(x_0, \dots, x_T) = (f(x_0), \dots, f(x_T))$, and the *observed path law* is

$$\mathbf{P}_T^{\mu,P,f} = f_{\#}^{(T)}\, \mathbf{P}_T^{\mu,P} \in \Delta(Y^{T+1}),$$

the law of the observed trajectory $(f(X_0), \dots, f(X_T))$. This is the only notion of observed process used in the paper. In general the observed process is not a Markov chain, and how dissipation appears under such aggregation is itself a subject of study [7].

2.3 Relative entropy and the arrow audit

For $p, q \in \Delta(\Omega)$ on a finite set Ω, the Kullback–Leibler divergence is

$$D_{\mathrm{KL}}(p \,\|\, q) = \sum_{\omega:\, p(\omega)>0} p(\omega) \log \frac{p(\omega)}{q(\omega)} \in [0, \infty],$$

with the value $+\infty$ whenever some ω has $p(\omega) > 0 = q(\omega)$. Terms with $p(\omega) = 0$ are omitted, so no product $0 \cdot \infty$ ever arises. The divergence is nonnegative and vanishes exactly when $p = q$ (Gibbs' inequality). For these standard facts see [3, 11].

Definition 2.1 (Arrow audits)**.** The *microscopic arrow* and the *observed arrow* at horizon T are

$$\mathcal{A}_T(\mu, P) = D_{\mathrm{KL}}\Big(\mathbf{P}_T^{\mu,P} \,\Big\|\, \mathrm{rev}_{\#}\mathbf{P}_T^{\mu,P}\Big), \qquad \mathcal{A}_T(\mu, P; f) = D_{\mathrm{KL}}\Big(\mathbf{P}_T^{\mu,P,f} \,\Big\|\, \mathrm{rev}_{\#}\mathbf{P}_T^{\mu,P,f}\Big).$$

Both values lie in $[0, \infty]$. An arrow is infinite when some path has positive probability but its reversal is impossible, as happens for a transition $x \to x'$ with $P(x, x') > 0 = P(x', x)$ that is used with positive probability. Forward-versus-reverse path divergences are the basic quantities of stochastic thermodynamics [14, 17]. The standard total entropy production starts the reversed paths from the terminal law μP^T, whereas the arrow reverses the path law itself; the two coincide for stationary chains, and in general they differ by a boundary term (Remark 4.3).

[1] The term comes from the literature on bidirectional transformations and the view-update problem [8]. Only the forward map is used here.

2.4 Graphs, gradient fields, and cycle affinity

Graphs are simple and undirected. For a graph $G = (V, E)$, an *edge field* is a function a that assigns a real number $a(u, v)$ to each ordered pair of adjacent vertices; it is *antisymmetric* if $a(v, u) = -a(u, v)$ for every edge $\{u, v\}$. An edge field is *exact*, or a *gradient*, if there is a potential $\phi : V \to \mathbb{R}$ with $a(u, v) = \phi(v) - \phi(u)$ for every edge $\{u, v\}$; only edges are constrained. The *sum* of a along a closed walk $v_0, v_1, \ldots, v_k = v_0$ (consecutive vertices adjacent) is $\sum_{i=0}^{k-1} a(v_i, v_{i+1})$. A *forest* is a graph without cycles, and a *tree* is a connected forest. For a finite graph with $c(G)$ connected components, the *cycle rank* $\beta_1(G) = |E| - |V| + c(G)$ is the dimension of its cycle space, and $\beta_1(G) = 0$ exactly for forests [1, 5]. Exactness and closed-walk sums are the degree-one part of discrete Hodge theory on graphs [13].

A Markov kernel P on X defines its *bidirected support graph* G_P on X: distinct x, x' are adjacent when $P(x, x') > 0$ and $P(x', x) > 0$. On G_P the *log-ratio field*

$$a_P(x, x') = \log \frac{P(x, x')}{P(x', x)}$$

is antisymmetric, and its sum around a cycle is the *cycle affinity* of Schnakenberg's network theory [16]. The sum of an antisymmetric field along a closed walk depends only on the walk's net flow through each edge, which is a real circulation; since the circulations of a finite graph are spanned by the circulations of any fundamental cycle basis, all closed-walk sums vanish if and only if they vanish on such a basis. Transitions with $P(x, x') > 0 = P(x', x)$ are *one-way*: they are not edges of G_P and have no finite log-ratio.

2.5 Closure deficit

Fix a kernel P on X, a lens $f : X \to Y$, a lag $\tau \geq 0$, and a law $\mu \in \Delta(X)$, which we call the *current law*. Let $(X_t)_{t \geq 0}$ be the chain started from $X_0 \sim \mu$, and write $Y_t = f(X_t)$. No stationarity is assumed. For each microstate x, its *packaged future law* is

$$p_x(y') = \sum_{z:\, f(z) = y'} P^\tau(x, z), \qquad y' \in Y.$$

It is defined for every x, and for $\mu(x) > 0$ it equals $\mathrm{Law}(Y_\tau \mid X_0 = x)$. A *macro kernel* is any Markov kernel K on Y; it predicts the macrostate at lag τ from the current macrostate alone. Its *closure loss* is

$$\mathcal{L}_\tau(K) = \sum_{x:\, \mu(x) > 0} \mu(x)\, D_{\mathrm{KL}}(p_x \,\|\, K(f(x), \cdot)) \in [0, \infty],$$

the average information lost by predicting each microstate's macroscopic future from its macrostate alone. The *closure deficit* is the conditional mutual information

$$\mathrm{CD}_\tau(\mu; P, f) = I(X_0; Y_\tau \mid Y_0),$$

computed from the joint law of (X_0, Y_0, Y_τ) [3]; it is always finite. It measures how much the present microstate tells about the future macrostate beyond what the present macrostate already tells. Exact preservation of the Markov property under aggregation is the classical lumpability problem [2, 10]; the closure deficit quantifies the failure of a weaker, one-lag property.

2.6 Total variation and Dobrushin's coefficient

For laws μ, ν on a finite set S, the total variation distance is

$$d_{\mathrm{TV}}(\mu,\nu) = \tfrac{1}{2}\sum_{s\in S}|\mu(s)-\nu(s)| = \max_{A\subseteq S}\bigl(\mu(A)-\nu(A)\bigr).$$

For a Markov kernel P on a nonempty finite set S, *Dobrushin's coefficient* is the largest total variation distance between two rows,

$$\lambda(P) = \max_{s,s'\in S} d_{\mathrm{TV}}\bigl(P(s,\cdot),P(s',\cdot)\bigr) \in [0,1].$$

It is also the smallest constant c such that the Markov operator $T_P(\nu) = \nu P$ satisfies $d_{\mathrm{TV}}(T_P\mu, T_P\nu) \le c\, d_{\mathrm{TV}}(\mu,\nu)$ for all laws μ,ν [6, 18]; we prove the inequality with $c = \lambda(P)$ in Lemma 7.1. A law ν is *ε-stable* if $d_{\mathrm{TV}}(\nu,\nu P) \le \varepsilon$; the 0-stable laws are the stationary laws.

2.7 Packaging and definable predicates

A *packaging map* is an endomap $e : S \to S$ of a set; it is *idempotent* if $e \circ e = e$. A *predicate* on a set X is a function $X \to \{0,1\}$. Given a lens $f : X \to Y$ between arbitrary sets, a predicate is *f-definable* if it factors through f:

$$\mathrm{Def}(f) = \{\, g \circ f \;:\; g : Y \to \{0,1\} \,\}.$$

An f-definable predicate cannot distinguish two microstates with the same image, so it is the indicator of a union of fibers of f. Predicates are ordered pointwise.

3 The eight theorems

This section states all eight theorems in one place, with a short reading of each. Proofs, examples, and further discussion follow in Sections 4 to 8.

Arrow of time

Theorem 3.1 (Arrow monotonicity)**.** *Let X and Y be finite sets, let $\mu \in \Delta(X)$, let P be a Markov kernel on X, let $f : X \to Y$ be a lens, and let $T \ge 0$. Then*

$$\mathcal{A}_T(\mu, P; f) \;\le\; \mathcal{A}_T(\mu, P),$$

that is,

$$D_{\mathrm{KL}}\Bigl(f^{(T)}_{\#}\mathbf{P}^{\mu,P}_T \,\Big\|\, \mathrm{rev}_{\#} f^{(T)}_{\#}\mathbf{P}^{\mu,P}_T\Bigr) \;\le\; D_{\mathrm{KL}}\Bigl(\mathbf{P}^{\mu,P}_T \,\Big\|\, \mathrm{rev}_{\#}\mathbf{P}^{\mu,P}_T\Bigr),$$

as an inequality in $[0,\infty]$.

The arrow seen through a lens never exceeds the arrow of the microscopic chain. No choice of lens, initial law, or horizon can make a coarse description look more irreversible than the system it describes. The inequality also holds when the microscopic arrow is infinite, in which case it says nothing; when the microscopic arrow is finite, so is every observed arrow.

Theorem 3.2 (Protocol trap)**.** *Let P be a Markov kernel on a finite set X, and let $\pi \in \Delta(X)$ satisfy detailed balance, $\pi(x)P(x,x') = \pi(x')P(x',x)$ for all $x, x' \in X$. Then for every horizon $T \geq 0$ the path law is reversal invariant,*

$$\mathrm{rev}_{\#}\mathbf{P}_T^{\pi,P} = \mathbf{P}_T^{\pi,P},$$

so $\mathcal{A}_T(\pi, P) = 0$. Consequently $\mathcal{A}_T(\pi, P; f) = 0$ for every lens $f : X \to Y$ into a finite set Y.

The observed process of a chain started in a detailed-balance law can be highly structured. It need not be Markov, it can have long memory, and hidden coordinates such as a protocol phase can impose a recognizable order on what is observed. It cannot be directional: every observed trajectory is exactly as likely as its reversal. Hidden protocol structure alone therefore does not produce an arrow; a positive observed arrow requires either a kernel without a detailed-balance law, which is a drive, or a start away from a detailed-balance law.

Forces and cycles

Theorem 3.3 (Forest exactness)**.** *Let $G = (V, E)$ be a forest, not necessarily finite, and let a be an antisymmetric edge field on G. Then a is exact: there is $\phi : V \to \mathbb{R}$ with*

$$a(u, v) = \phi(v) - \phi(u) \qquad \text{for every edge } \{u, v\} \in E.$$

Theorem 3.4 (Gradient fields have no cycle affinity)**.** *Let $G = (V, E)$ be a graph, let $\phi : V \to \mathbb{R}$, and let a be an edge field with $a(u, v) = \phi(v) - \phi(u)$ for every edge $\{u, v\} \in E$. Then for every closed walk $v_0, v_1, \ldots, v_k = v_0$ in G,*

$$\sum_{i=0}^{k-1} a(v_i, v_{i+1}) = 0.$$

Together these two theorems locate force-like structure. A field that is a gradient has zero sum around every loop, and on a graph without loops every antisymmetric field is a gradient. For the log-ratio field a_P of a Markov kernel, a nonzero cycle affinity therefore needs two things: a cycle in the bidirected support graph, and transition rates whose log-ratios do not come from a potential. Neither thresholding a graph nor regularizing its rates is covered: both change the graph or the field, and so change the object the theorems are about.

Closure of macroscopic laws

Theorem 3.5 (Closure deficit)**.** *Let P be a Markov kernel on a finite set X, let $f : X \to Y$ be a lens into a finite set, let $\tau \geq 0$, and let $\mu \in \Delta(X)$ be the current law. Define $K^\star$ by*

$$K^\star(y, \cdot) = \mathrm{Law}(Y_\tau \mid Y_0 = y) = \frac{1}{\mu(f^{-1}(y))} \sum_{x \in f^{-1}(y)} \mu(x)\, p_x \qquad \text{if } \mu(f^{-1}(y)) > 0,$$

and let $K^\star(y, \cdot)$ be an arbitrary law if $\mu(f^{-1}(y)) = 0$. Then:

(i) $\mathcal{L}_\tau(K^\star) = \mathrm{CD}_\tau(\mu; P, f)$*;*

(ii) $\mathrm{CD}_\tau(\mu; P, f) \leq \mathcal{L}_\tau(K)$ *for every macro kernel K, including kernels with $\mathcal{L}_\tau(K) = \infty$; hence* $\mathrm{CD}_\tau(\mu; P, f) = \min_K \mathcal{L}_\tau(K)$*;*

(iii) $\mathrm{CD}_\tau(\mu; P, f) = 0$ *if and only if* $p_x = p_{x'}$ *whenever* $f(x) = f(x')$, $\mu(x) > 0$ *and* $\mu(x') > 0$. *In particular, if two microstates of positive mass in the same fiber have different packaged future laws, then* $\mathrm{CD}_\tau(\mu; P, f) > 0$.

The theorem has three parts. The best macro kernel is the obvious one: from each macrostate, predict the average future of the microstates it contains, weighted by their current mass. The loss that even the best kernel incurs is a conditional mutual information, so it has an information-theoretic meaning independent of any fitting procedure. This loss vanishes exactly when the macrostate already carries all the information about the future macrostate that the microstate does, which can be certified or refuted by comparing future laws within fibers. The current law matters: the same kernel, lens, and lag can have positive deficit for one law and zero deficit for another (Example 6.1).

Objects under contraction

Theorem 3.6 (Contractive objecthood)**.** *Let P be a Markov kernel on a nonempty finite set S with Dobrushin coefficient $\lambda(P) < 1$, and let $\varepsilon \geq 0$. If $\nu, \nu' \in \Delta(S)$ are ε-stable, that is, $d_{\mathrm{TV}}(\nu, \nu P) \leq \varepsilon$ and $d_{\mathrm{TV}}(\nu', \nu' P) \leq \varepsilon$, then*

$$d_{\mathrm{TV}}(\nu, \nu') \;\leq\; \frac{2\varepsilon}{1 - \lambda(P)}.$$

In particular, P has at most one stationary law.

Read an ε-stable law as a candidate object: a distribution that the dynamics changes by at most ε in one step. Under strict contraction all candidate objects lie within a common distance $2\varepsilon/(1 - \lambda(P))$ of each other, so the dynamics cannot carry two robust objects that are farther apart; for $\varepsilon > 0$ and at least two states there are infinitely many candidate objects, since every law within $\varepsilon/(1 + \lambda(P))$ of the stationary law is ε-stable, but they all crowd together. Equivalently, two ε-stable laws at distance $d > 0$ force $\lambda(P) \geq 1 - 2\varepsilon/d$, and two distinct stationary laws force $\lambda(P) = 1$.

Ladders of levels

Theorem 3.7 (Idempotent saturation)**.** *Let S be any set and let $e : S \to S$ satisfy $e \circ e = e$. Then $e^{\circ n} = e$ for every integer $n \geq 1$.*

Theorem 3.8 (Bounded interface)**.** *Let $f : X \to Y$ be a map between arbitrary sets whose image* $\mathrm{im}(f)$ *is finite. Then*

$$|\mathrm{Def}(f)| = 2^{|\mathrm{im}(f)|}.$$

Consequently there is no infinite sequence $h_1, h_2, \ldots$ of pairwise distinct f-definable predicates; in particular there is no infinite strictly increasing chain of them.

A ladder is a sequence of levels in which each level adds something the previous ones did not have. Packaging the output of an idempotent packaging map again adds nothing. Observation through a fixed interface with finite image can tell apart only finitely many properties, so any sequence of genuinely new distinctions drawn from it must stop. An unbounded ladder requires changing the rules: a packaging map that is not idempotent, or a sequence of interfaces in place of a single fixed one (Remark 8.2).

4 Arrow of time under observation

This section proves Theorem 3.1 and Theorem 3.2. Both rest on two standard facts: relative entropy cannot increase under data processing [3], and a chain in detailed balance looks the same forwards and backwards [9]. For a stationary chain the arrow is the entropy production over the observation window [14, 17], and the results then say that coarse observation can hide dissipation but never create it [7].

4.1 Observation cannot create an arrow

Lemma 4.1 (Deterministic data processing)**.** *Let $g : \Omega \to \Omega'$ be a map between finite sets. For all $p, q \in \Delta(\Omega)$,*

$$D_{\mathrm{KL}}(g_{\#}p \,\|\, g_{\#}q) \le D_{\mathrm{KL}}(p \,\|\, q) \qquad \text{in } [0, \infty].$$

Proof. See Appendix A; the proof applies the log-sum inequality fiber by fiber. □

Lemma 4.2 (Reversal commutes with observation)**.** *For every $T \ge 0$, $f^{(T)} \circ \mathrm{rev} = \mathrm{rev} \circ f^{(T)}$ as maps $X^{T+1} \to Y^{T+1}$. Consequently, for every $Q \in \Delta(X^{T+1})$,*

$$f^{(T)}_{\#}(\mathrm{rev}_{\#}Q) = \mathrm{rev}_{\#}(f^{(T)}_{\#}Q).$$

Proof. Both composites send $(x_0, \dots, x_T)$ to $(f(x_T), \dots, f(x_0))$. Pushforward respects composition, so $f^{(T)}_{\#}\mathrm{rev}_{\#}Q = (f^{(T)} \circ \mathrm{rev})_{\#}Q = (\mathrm{rev} \circ f^{(T)})_{\#}Q = \mathrm{rev}_{\#}f^{(T)}_{\#}Q$. □

Proof of Theorem 3.1. Write $Q = \mathbf{P}_T^{\mu,P}$. By Lemma 4.2, the observed path law and its reversal are the images of Q and $\mathrm{rev}_{\#}Q$ under the single map $g = f^{(T)}$:

$$\mathbf{P}_T^{\mu,P,f} = g_{\#}Q, \qquad \mathrm{rev}_{\#}\mathbf{P}_T^{\mu,P,f} = g_{\#}(\mathrm{rev}_{\#}Q).$$

Lemma 4.1 with $p = Q$ and $q = \mathrm{rev}_{\#}Q$ gives $\mathcal{A}_T(\mu, P; f) \le \mathcal{A}_T(\mu, P)$. □

The only property of the observation used in the proof is that it commutes with reversal. This identifies which observations are covered and which are not.

Remark 4.1 (Noisy observation is covered; time-dependent observation is not)**.** Suppose each X_t is observed through the same noisy channel C, a Markov kernel from X to Y, independently at each time. The observed path law is then the image of $\mathbf{P}_T^{\mu,P}$ under the product channel $C^{\otimes(T+1)}$. That channel commutes with reversal, and the data-processing inequality holds for arbitrary channels [3], so the conclusion of Theorem 3.1 still holds. Leaving the deterministic setting is therefore not by itself a way to create an arrow.

An observation that depends on time is different. Let X have a single state, so that every arrow of the chain is zero, and record 0 at time 0 and 1 at time 1. The observed path is $(0, 1)$ with probability one and its reversal $(1, 0)$ has probability zero, so the observed arrow is infinite. Here the arrow comes from a clock in the observer, not from the system. A Markov chain fitted to the observed process is a third kind of object; the theorem makes no claim about it.

4.2 The protocol trap

Lemma 4.3 (Detailed balance gives reversible paths)**.** *If π satisfies detailed balance with P, then for every $T \geq 0$ and every path $(x_0, \ldots, x_T)$,*

$$\pi(x_0) \prod_{t=0}^{T-1} P(x_t, x_{t+1}) = \pi(x_T) \prod_{t=0}^{T-1} P(x_{t+1}, x_t).$$

Equivalently, $\mathbf{P}_T^{\pi,P} = \mathrm{rev}_{\#}\mathbf{P}_T^{\pi,P}$.

Proof. See Appendix A. The proof is an induction on T that uses detailed balance once per step and never divides, so it covers states with $\pi(x) = 0$. □

Proof of Theorem 3.2. By Lemma 4.3 the path law equals its reversal, so

$$\mathcal{A}_T(\pi, P) = D_{\mathrm{KL}}\Big(\mathbf{P}_T^{\pi,P} \,\Big\|\, \mathbf{P}_T^{\pi,P}\Big) = 0.$$

By Theorem 3.1, $0 \leq \mathcal{A}_T(\pi, P; f) \leq \mathcal{A}_T(\pi, P) = 0$ for every lens f. □

Remark 4.2 (What the protocol trap does and does not rule out)**.** The theorem is about the arrow audit and nothing else. It allows the observed process $f(X_t)$ to have any amount of temporal structure, and in particular it allows a hidden protocol coordinate to make the observed process pass through recognizable stages. What it rules out is a preferred direction: under a detailed-balance start, every observed sequence of stages is exactly as likely as the reverse sequence. The hypothesis concerns the pair (π, P) and not P alone, as the next example shows.

Example 4.1 (A reversible kernel with a nonequilibrium start)**.** Let $X = \{1, 2\}$ and let P have all entries equal to $1/2$. The uniform law satisfies detailed balance, and started from it the chain has zero arrow. Started instead from $\mu = \delta_1$, the path $(1, 2)$ has probability $1/2$ while its reversal $(2, 1)$ has probability 0, so $\mathcal{A}_1(\delta_1, P) = \infty$. The arrow here records the asymmetry of the initial condition, not a drive: in the decomposition of Remark 4.3, the total entropy production is $\log 2$ and the boundary term $D_{\mathrm{KL}}(\mu P \,\|\, \mu)$ is infinite. A constant lens hides the arrow completely, while the identity lens shows all of it.

Remark 4.3 (Relation to entropy production)**.** Write $\mu_T = \mu P^T$ for the law at time T. The standard total entropy production over the window compares the forward path law with the reversal of the path law started from μ_T [14, 17]:

$$\Sigma_T(\mu, P) = D_{\mathrm{KL}}\Big(\mathbf{P}_T^{\mu,P} \,\Big\|\, \mathrm{rev}_{\#}\mathbf{P}_T^{\mu_T,P}\Big)\,.$$

The two quantities are related by

$$\mathcal{A}_T(\mu, P) = \Sigma_T(\mu, P) + D_{\mathrm{KL}}(\mu_T \,\|\, \mu) \qquad \text{in } [0, \infty].$$

Indeed, the two reversed laws give a path $(x_0, \ldots, x_T)$ the masses $\mu(x_T)B$ and $\mu_T(x_T)B$ with the same backward product $B = \prod_t P(x_{t+1}, x_t)$. If all three terms are finite, splitting the logarithm of the likelihood ratio and using that x_T has law μ_T under the forward path law gives the identity. Otherwise both sides are infinite: the left side is infinite exactly when some path of positive probability has $\mu(x_T) = 0$ or $B = 0$; since its endpoint has $\mu_T(x_T) > 0$, the first case is exactly the case $D_{\mathrm{KL}}(\mu_T \,\|\, \mu) = \infty$ and the second the case $\Sigma_T = \infty$. Both reversed laws commute with the lens

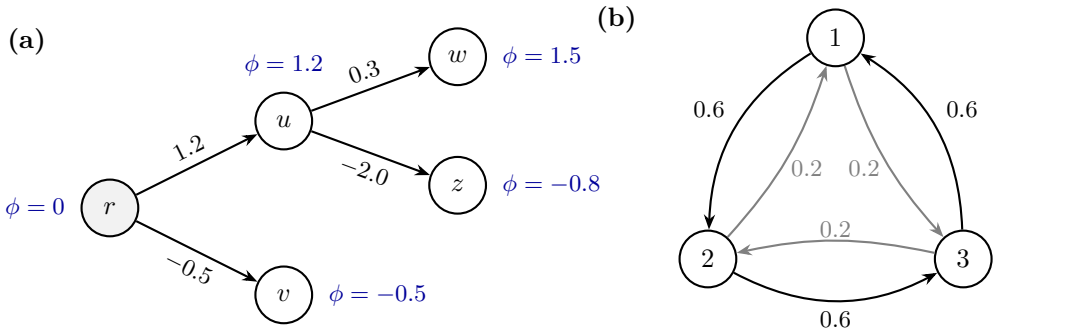


Figure 2. (a) On a tree every antisymmetric edge field is a gradient. Edge labels give $a(\text{parent}, \text{child})$; summing them along the unique path from the root r gives the potential ϕ (blue), and $a(u, v) = \phi(v) - \phi(u)$ on every edge. (b) A driven three-state ring, with holding probability 0.2 at each state. Each log-ratio equals $\log 3$, so the cycle affinity is $3 \log 3$ and the field is not a gradient. Deleting any one edge of the ring leaves a tree, on which the restricted field is a gradient again.

in the same way, so the proof of Theorem 3.1 also gives $D_{\mathrm{KL}}\big(f_{\#}^{(T)}\mathbf{P}_T^{\mu,P} \,\big\|\, \mathrm{rev}_{\#} f_{\#}^{(T)}\mathbf{P}_T^{\mu_T,P}\big) \le \Sigma_T(\mu, P)$: observation cannot increase the total entropy production either.

If π is stationary, the boundary term vanishes, and for $T \ge 1$

$$\mathcal{A}_T(\pi, P) = \Sigma_T(\pi, P) = T\,\sigma(\pi, P), \qquad \sigma(\pi, P) = \sum_{x,x'} \pi(x)P(x, x') \log \frac{\pi(x)P(x, x')}{\pi(x')P(x', x)},$$

where σ is the stationary entropy production rate [16, 17], the sum runs over pairs with $\pi(x)P(x, x') > 0$, and $\sigma = +\infty$ if some such pair has $P(x', x) = 0$. For a stationary chain, Theorem 3.1 thus bounds the observed arrow by $T\sigma(\pi, P)$.

5 Forces need cycles

This section proves Theorem 3.3 and Theorem 3.4 and connects them to Markov kernels. The graph-theoretic content is classical [1, 5]: it is the statement that a 1-form on a graph is exact precisely when it integrates to zero around every cycle, specialized to the two directions we need [13]. In Markov network theory the cycle sums of the log-ratio field are the affinities that drive steady-state currents [16]. Figure 2 illustrates both theorems.

5.1 Proofs

Proof of Theorem 3.3. In each connected component of G choose a root. For a vertex v, let $r = v_0, v_1, \dots, v_m = v$ be the path in G from the root of its component to v; it exists because the component is connected and is unique because G has no cycles. Define

$$\phi(v) = \sum_{i=0}^{m-1} a(v_i, v_{i+1}),$$

so the root has $\phi(r) = 0$. Appendix B checks that $a(u, v) = \phi(v) - \phi(u)$ on every edge. Paths are finite even if G is infinite, so finiteness of G is never used. □

Proof of Theorem 3.4. Along a closed walk $v_0, \dots, v_k = v_0$ the sum telescopes:

$$\sum_{i=0}^{k-1} a(v_i, v_{i+1}) = \sum_{i=0}^{k-1} \big(\phi(v_{i+1}) - \phi(v_i)\big) = \phi(v_k) - \phi(v_0) = 0.$$

Only the values of a on edges enter, and every step of a walk is an edge. □

The converse of Theorem 3.4 also holds: on a connected graph, an antisymmetric field whose sums vanish around every closed walk is a gradient, with potential obtained by summing a along any path from a fixed root. Theorem 3.3 is the special case in which there are no cycles to check.

5.2 Consequences for Markov kernels

Apply the two theorems to the log-ratio field a_P on the bidirected support graph G_P. If G_P is a forest, then a_P is a gradient; if a_P is a gradient, then every cycle affinity vanishes. A nonzero cycle affinity therefore requires both a cycle in G_P and a log-ratio field that is not a gradient. For chains whose transitions all have reverses, forest support has a stronger consequence, which connects this section to the protocol trap.

Corollary 5.1 (Forest-supported chains are reversible)**.** *Let P be a Markov kernel on a nonempty finite set X without one-way transitions, that is, $P(x, x') > 0$ if and only if $P(x', x) > 0$ for all $x \neq x'$. If G_P is a forest, then P has a detailed-balance law π with $\pi(x) > 0$ for all x. Consequently $\mathcal{A}_T(\pi, P; f) = 0$ for every lens f and every horizon T.*

Proof. By Theorem 3.3, $a_P(x, x') = \phi(x') - \phi(x)$ on every edge of G_P for some ϕ. Put $\pi(x) = e^{\phi(x)}/Z$ with $Z = \sum_x e^{\phi(x)}$. On an edge, $P(x, x')/P(x', x) = e^{\phi(x') - \phi(x)}$, which rearranges to $\pi(x)P(x, x') = \pi(x')P(x', x)$. For distinct non-adjacent x, x' both sides vanish because there are no one-way transitions, and for $x = x'$ the identity is trivial. So π satisfies detailed balance, and Theorem 3.2 gives the last claim. □

This is the finite form of the classical fact that a Markov chain whose transition graph is a tree is reversible [9].

Remark 5.1 (Gradient log-ratios do not imply detailed balance)**.** The hypothesis on one-way transitions in Corollary 5.1 cannot be dropped. One-way transitions are not edges of G_P, so a_P does not see them, yet a transition with $P(x, x') > 0 = P(x', x)$ forces every detailed-balance law to vanish at x. For instance, the kernel on $\{1, 2\}$ with $P(1, 2) = 1$ and $P(2, 2) = 1$ has an empty bidirected support graph, so a_P is trivially a gradient, but the only detailed-balance law is δ_2; started from δ_1, the chain has infinite arrow. An audit for the protocol trap must therefore check detailed balance on all pairs of states, not only exactness of the log-ratio field.

Remark 5.2 (Additive affinities and multiplicative surrogates)**.** The cycle affinity is additive: it is the logarithm of the product r of rate ratios around the cycle. Numerical audits sometimes report a multiplicative surrogate such as $\max(r, 1/r) - 1 = e^{|\log r|} - 1$. It vanishes exactly when the affinity does and orders cycles by $|\log r|$, but it is a different number from the affinity.

Remark 5.3 (Thresholded and regularized graphs)**.** Both theorems are statements about the exact graph and the exact field. Discarding small transition probabilities removes edges, can turn a cycle into a forest, and can create one-way transitions; adding a small floor to all probabilities adds edges, can create cycles, and replaces one-way transitions by edges with large but finite log-ratios. Either operation produces a different graph and a different field, about which the theorems say nothing.

6 Closure of macroscopic laws

A lens f turns a Markov chain into an observed process $Y_t = f(X_t)$ that is usually not Markov. One may still ask for the best Markov description at a fixed lag: a kernel K on macrostates that predicts Y_τ from Y_0. Theorem 3.5 identifies the best such kernel and the loss it cannot avoid. Throughout this section P, f, τ, and the current law μ are fixed, and p_x, $\mathcal{L}_\tau$, and $\mathrm{CD}_\tau = \mathrm{CD}_\tau(\mu; P, f)$ are as in Section 2. For a macrostate y write $W_y = \mu(f^{-1}(y))$ for the current mass of its fiber. The minimization below is a finite instance of information projection [4], and the identities are standard finite calculations with conditional mutual information [3].

6.1 Three lemmas and the proof

Lemma 6.1 (Mixture decomposition)**.** *Let $w_1, \dots, w_n \geq 0$ have positive sum W, let $p_1, \dots, p_n \in \Delta(\Omega)$, and let $\bar{p} = W^{-1} \sum_i w_i p_i$. Then for every $q \in \Delta(\Omega)$,*

$$\sum_{i:\, w_i > 0} w_i \, D_{\mathrm{KL}}(p_i \,\|\, q) = \sum_{i:\, w_i > 0} w_i \, D_{\mathrm{KL}}(p_i \,\|\, \bar{p}) + W \, D_{\mathrm{KL}}(\bar{p} \,\|\, q) \qquad \text{in } [0, \infty],$$

and the first sum on the right is finite. In particular $q = \bar{p}$ minimizes the left-hand side, and it is the unique minimizer.

Proof. See Appendix C. The proof treats the case where q misses part of the support of $\bar{p}$, in which both sides are infinite, separately from the finite case, so that no difference of infinite quantities is ever formed. □

Lemma 6.2 (Best macro kernel)**.** *For every macro kernel K, $\mathcal{L}_\tau(K^\star) \leq \mathcal{L}_\tau(K)$, and $\mathcal{L}_\tau(K^\star)$ is finite. If $W_y > 0$, then $K^\star(y, \cdot)$ is the only row that attains the minimum of the fiber's contribution; rows of fibers with $W_y = 0$ do not affect $\mathcal{L}_\tau$.*

Proof. See Appendix C. The objective splits into one term per fiber, and Lemma 6.1 minimizes each term with positive mass separately. □

Lemma 6.3 (Information identity)**.** $\mathrm{CD}_\tau = I(X_0; Y_\tau \mid Y_0) = \sum_{x:\, \mu(x) > 0} \mu(x) \, D_{\mathrm{KL}}(p_x \,\|\, K^\star(f(x), \cdot)) = \mathcal{L}_\tau(K^\star)$.

Proof. See Appendix C. Because $Y_0 = f(X_0)$ is a function of X_0, the conditional mutual information collapses to an average over microstates of positive mass. □

Proof of Theorem 3.5. The first displayed formula for $K^\star$ is the definition of a conditional law: for $W_y > 0$, $\Pr(Y_0 = y, Y_\tau = y') = \sum_{x \in f^{-1}(y)} \mu(x) \, p_x(y')$. Part (i) is Lemma 6.3, and part (ii) follows from (i) and Lemma 6.2. For part (iii), every term of the sum in Lemma 6.3 is finite and nonnegative, so $\mathrm{CD}_\tau = 0$ if and only if $p_x = K^\star(f(x), \cdot)$ for every x with $\mu(x) > 0$, by Gibbs' inequality. If this holds, then all positive-mass microstates in one fiber share the law $K^\star(y, \cdot)$. Conversely, if all positive-mass microstates in the fiber of y share one law, that law equals their weighted average $K^\star(y, \cdot)$. □

6.2 The current law matters

The deficit depends on the law of the present microstate, and the following example shows that the dependence is real.

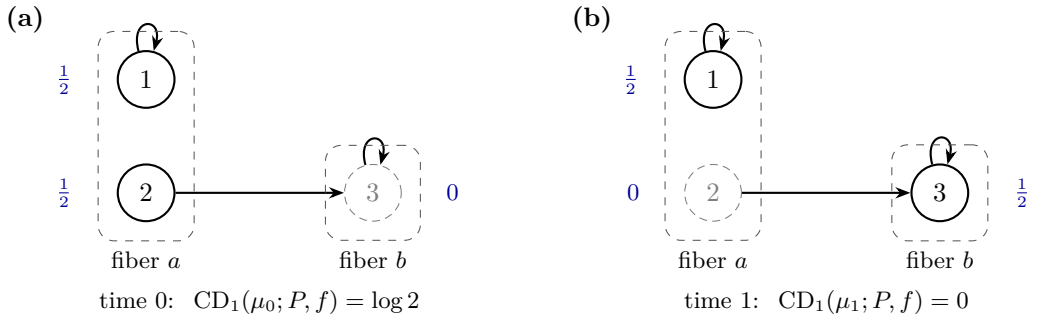


Figure 3. The closure deficit depends on the current law (Example 6.1). Microstates 1 and 2 form macrostate a, and state 3 forms macrostate b; current masses are shown in blue, and states of zero mass are dashed. (a) At time 0 both microstates in fiber a have mass $1/2$, but 1 stays in a while 2 moves to b, so the macrostate does not determine the macroscopic future and $\mathrm{CD}_1 = \log 2$. (b) One step later state 2 has mass zero, every positive-mass fiber has a single macroscopic future, and $\mathrm{CD}_1 = 0$.

Example 6.1 (Positive deficit now, zero deficit one step later)**.** Let $X = \{1, 2, 3\}$ and $Y = \{a, b\}$, with $f(1) = f(2) = a$ and $f(3) = b$, and let P be deterministic with $1 \mapsto 1$, $2 \mapsto 3$, $3 \mapsto 3$. Take $\tau = 1$, so $p_1 = \delta_a$ and $p_2 = p_3 = \delta_b$ (Figure 3).

- For $\mu_0 = (\frac{1}{2}, \frac{1}{2}, 0)$, the two states of fiber a have positive mass and different futures, so $\mathrm{CD}_1(\mu_0) > 0$. Indeed $K^\star(a, \cdot) = (\frac{1}{2}, \frac{1}{2})$, $Y_0 = a$ almost surely, and Y_1 is a fair coin determined by X_0, so $\mathrm{CD}_1(\mu_0) = \log 2$.
- For $\mu_1 = \mu_0 P = (\frac{1}{2}, 0, \frac{1}{2})$, each fiber contains one state of positive mass, so $\mathrm{CD}_1(\mu_1) = 0$.

The kernel, lens, and lag are the same in both cases; only the current law differs.

For a chain started from μ, the deficit between times t and $t + \tau$ is therefore $\mathrm{CD}_\tau(\mu P^t; P, f)$, and the theorem applies at time t with the law μP^t. The objective must always be weighted by the law at the time from which the prediction is made: weighting the losses at time t by the initial law μ instead of μP^t gives a quantity that is not the conditional mutual information at time t. In the example, the packaged future laws do not depend on time, so the μ_0-weighted loss has minimum $\log 2$ at every time, while the deficit at time 1 is zero. If $\mu = \pi$ is stationary, then $\mu P^t = \pi$ for all t and the deficit is the same at every time.

6.3 Scope

Remark 6.1 (Relation to lumpability)**.** The kernel P is *strongly lumpable* for f if $P(x, f^{-1}(y')) = P(x', f^{-1}(y'))$ whenever $f(x) = f(x')$, for all states and not only those of positive mass [2, 10]. Strong lumpability then also holds for P^τ, so all packaged future laws agree within fibers and $\mathrm{CD}_\tau(\mu; P, f) = 0$ for every μ and τ. The converse fails: a zero deficit concerns one current law and one lag, and it constrains only the microstates of positive mass under that law.

Remark 6.2 (What the theorem does not decide)**.** The theorem certifies a positive deficit from a single pair of microstates of positive mass in one fiber whose macroscopic futures differ. It does not say which feature of the model produced that pair, nor whether a different lens, lag, or current law would remove it. A first-order Markov chain fitted to the observed process can be a useful summary, but its fitting error is not the closure deficit unless it is computed as above, with the current law as weights.

7 Objects under contraction

This section proves Theorem 3.6. Throughout, S is a nonempty finite set, P a Markov kernel on S, and $\lambda = \lambda(P)$ its Dobrushin coefficient. Dobrushin's coefficient and its use for the ergodicity of finite chains are classical [6, 18].

7.1 Contraction and the separation bound

Lemma 7.1 (Dobrushin contraction)**.** *For all $\mu, \nu \in \Delta(S)$, $d_{\mathrm{TV}}(\mu P, \nu P) \le \lambda(P)\, d_{\mathrm{TV}}(\mu, \nu)$.*

Proof. See Appendix A. For an event A, the proof compares each row's probability of A with the smallest such probability among all rows; this makes every coefficient in the resulting sum lie between 0 and $\lambda(P)$. □

Proof of Theorem 3.6. By the triangle inequality and Lemma 7.1,

$$d_{\mathrm{TV}}(\nu, \nu') \le d_{\mathrm{TV}}(\nu, \nu P) + d_{\mathrm{TV}}(\nu P, \nu' P) + d_{\mathrm{TV}}(\nu' P, \nu') \le 2\varepsilon + \lambda\, d_{\mathrm{TV}}(\nu, \nu').$$

All quantities are finite, so $(1 - \lambda)\, d_{\mathrm{TV}}(\nu, \nu') \le 2\varepsilon$, and dividing by $1 - \lambda > 0$ gives the bound. If ν and ν' are both stationary, take $\varepsilon = 0$: then $d_{\mathrm{TV}}(\nu, \nu') = 0$, so $\nu = \nu'$. □

The theorem bounds the distance between candidate objects but does not assert that any exists. Existence, together with convergence to the stationary law, follows from the contraction principle.

Proposition 7.1 (Existence and geometric convergence)**.** *If $\lambda < 1$, then P has a unique stationary law π, and for every $\mu \in \Delta(S)$ and $t \ge 0$,*

$$d_{\mathrm{TV}}(\mu P^t, \pi) \le \lambda^t\, d_{\mathrm{TV}}(\mu, \pi).$$

Proof. See Appendix A. Existence of a stationary law holds for every kernel on a nonempty finite set; the proof in the appendix obtains it from the contraction instead, as the limit of the iterates μP^t. □

Corollary 7.1 (All candidate objects lie near the stationary law)**.** *If $\lambda < 1$ and ν is ε-stable, then $d_{\mathrm{TV}}(\nu, \pi) \le \varepsilon/(1 - \lambda)$.*

Proof. $d_{\mathrm{TV}}(\nu, \pi) \le d_{\mathrm{TV}}(\nu, \nu P) + d_{\mathrm{TV}}(\nu P, \pi P) \le \varepsilon + \lambda\, d_{\mathrm{TV}}(\nu, \pi)$; rearrange. □

Figure 4 shows these bounds for a kernel on three states.

7.2 Scope

Remark 7.1 (Contraction is sufficient, not necessary)**.** A unique stationary law does not require $\lambda < 1$. The kernel that swaps two states has $\lambda = 1$ and the unique stationary law $(\frac{1}{2}, \frac{1}{2})$, although its iterates do not converge. Theorem 3.6 gives a quantitative bound only in the strictly contractive regime.

Remark 7.2 (Stationary laws are not fixed states)**.** For a deterministic map, the number of fixed states and the number of stationary laws are different counts. The map on $\{1, 2, 3\}$ with $1 \mapsto 1$, $2 \mapsto 3$, $3 \mapsto 2$ has one fixed state but two extreme stationary laws, δ_1 and $(0, \frac{1}{2}, \frac{1}{2})$, and hence infinitely many stationary laws, their mixtures. Its Dobrushin coefficient is 1, consistent with the theorem. An audit of objecthood must count stationary laws, not fixed states.

Remark 7.3 (Metric)**.** The theorem uses total variation together with Dobrushin's coefficient. Contraction in another metric, or approximate stability measured differently, needs its own argument. Clustering heuristics that group laws by a threshold on some other distance are not covered.

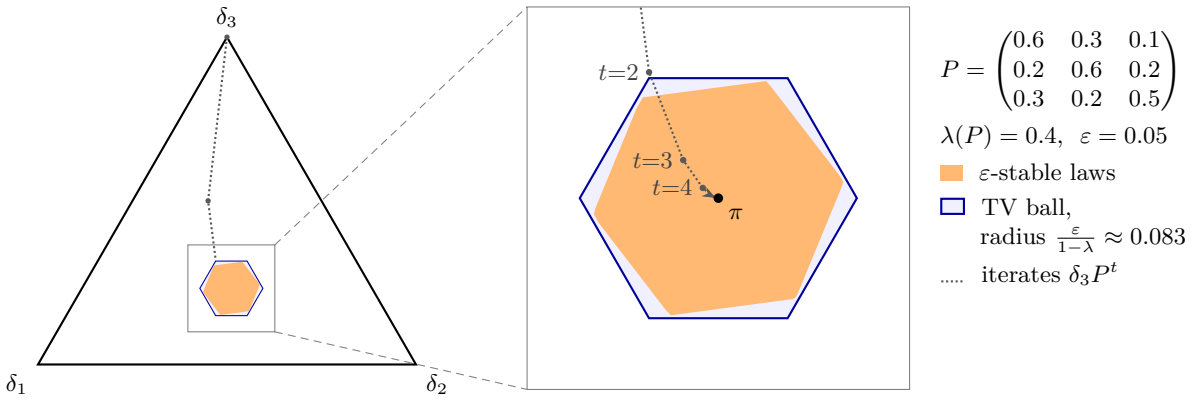


Figure 4. Contractive objecthood on three states. Left: the simplex $\Delta(\{1,2,3\})$; the boxed region is enlarged on the right. The set of ε-stable laws (orange, computed on a fine grid) lies inside the total variation ball of radius $\varepsilon/(1-\lambda)$ around the stationary law π guaranteed by Corollary 7.1; in the plane of the simplex this ball is a hexagon. Any two orange laws are therefore within $2\varepsilon/(1-\lambda) \approx 0.17$ of each other, as Theorem 3.6 asserts. The dotted path shows the iterates $\delta_3 P^t$ converging geometrically to π (Proposition 7.1).

8 Ladders of levels

A ladder is a sequence of levels, each adding a distinction that the previous ones lacked. This section proves the two theorems that bound ladders built from a fixed packaging map or a fixed interface. Both are elementary; their content lies in what they force an unbounded ladder to change.

8.1 Idempotent packaging saturates

Lemma 8.1. *If $e \circ e = e$, then $e^{\circ n} = e$ for all $n \geq 1$.*

Proof. By induction on n; see Appendix D. □

Proof of Theorem 3.7. This is Lemma 8.1. In particular, any quantity that depends on a state only through its packaged value $e^{\circ n}(s)$ takes the same value for all $n \geq 1$. □

8.2 A fixed interface has finitely many definable predicates

Lemma 8.2. *Let $f : X \to Y$ be any map. The assignment $A \mapsto \mathbf{1}_A \circ f$ is a bijection from the subsets of* $\mathrm{im}(f)$ *onto* $\mathrm{Def}(f)$*. It is an order isomorphism: $\mathbf{1}_A \circ f \leq \mathbf{1}_B \circ f$ pointwise if and only if $A \subseteq B$.*

Proof. See Appendix D. Only the values of g on the image of f affect $g \circ f$, and every point of the image is hit by some x. □

Proof of Theorem 3.8. By Lemma 8.2, $|\mathrm{Def}(f)|$ is the number of subsets of $\mathrm{im}(f)$, which is $2^{|\mathrm{im}(f)|}$. An infinite sequence of pairwise distinct elements would make the finite set $\mathrm{Def}(f)$ infinite, which is impossible. A strictly increasing chain consists of pairwise distinct elements, so none is infinite either. □

Figure 5 illustrates the bijection. The order isomorphism gives a sharper bound on chains.

Remark 8.1 (Length of chains)**.** If $|\mathrm{im}(f)| = m$, every strictly increasing chain of f-definable predicates has at most $m+1$ elements, and chains of exactly $m+1$ elements exist. By the order

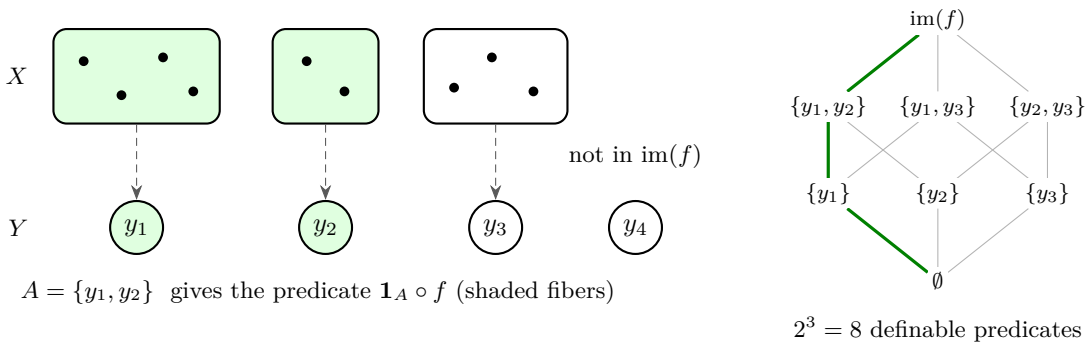


Figure 5. Definable predicates of a lens with $|\mathrm{im}(f)| = 3$. Left: a predicate definable through f is constant on fibers, so it is determined by the set $A \subseteq \mathrm{im}(f)$ of images on which it equals one; values of g off the image, such as at y_4, are irrelevant. Right: the 8 definable predicates ordered pointwise form the lattice of subsets of $\mathrm{im}(f)$. A strictly increasing chain, such as the highlighted one, has at most $|\mathrm{im}(f)| + 1 = 4$ elements.

isomorphism, such a chain is a strictly increasing chain of subsets of an m-element set, and each step adds at least one element. Adding one element at a time from $\emptyset$ to $\mathrm{im}(f)$ gives a chain of length $m + 1$.

Remark 8.2 (How ladders can continue)**.** Both theorems fix the rules of the game. An unbounded ladder requires changing them: a packaging map that is not idempotent, or a sequence of interfaces $f_1, f_2, \ldots$ in place of a single fixed one. A changing interface need not have a growing image. On $X = \mathbb{N}$ the interfaces $f_n = \mathbf{1}_{\{0,\ldots,n\}}$ all have a two-point image, yet the predicates $f_1 < f_2 < \cdots$, each definable through its own interface, form an infinite strictly increasing chain. Growth becomes necessary when the interfaces are successive refinements, $f_n = g_n \circ f_{n+1}$ for maps g_n, as when coordinates are added or the lens is refined. Then $\mathrm{Def}(f_n) \subseteq \mathrm{Def}(f_{n+1})$, and g_n maps $\mathrm{im}(f_{n+1})$ onto $\mathrm{im}(f_n)$, so $|\mathrm{im}(f_n)|$ is nondecreasing. If these sizes stay bounded, they are eventually constant; from then on each g_n restricts to a bijection $\mathrm{im}(f_{n+1}) \to \mathrm{im}(f_n)$, every predicate definable through f_{n+1} is already definable through f_n, and the union of all $\mathrm{Def}(f_n)$ is a finite set. Along a chain of refinements, an infinite sequence of new distinctions therefore requires $|\mathrm{im}(f_n)| \to \infty$. In the vocabulary of the introduction, such growth is a form of staging or refinement.

9 Discussion

What the theorems say together. Each theorem fixes an exact audit and names a hypothesis under which the audit cannot show the emergent feature. Read the other way, each one names something that an emergence claim must supply. An arrow of time seen through a lens must already be present in the microscopic path law, either through a drive or through an initial law out of detailed balance; hidden protocol coordinates do not supply it. A force-like cycle affinity needs a cycle in the bidirected support and a log-ratio field that is not a gradient. A closed macroscopic law at a given lag exists, at zero loss, exactly when the positive-mass microstates of each macrostate agree about the macroscopic future, and the loss of the best macroscopic law otherwise has an exact information-theoretic value. Two distinct stationary laws require a dynamics that is not strictly contractive, and two ε-stable laws farther apart than $2\varepsilon/(1 - \lambda(P))$ are impossible when $\lambda(P) < 1$. An unbounded ladder of levels requires a packaging map that is not idempotent or a changing interface, and along a chain of successive refinements the interface must grow without bound. None of these requirements is surprising on its own. What the theorems add is precision about the object

being audited, so that a claim of emergence can be checked against an exact quantity rather than an approximation of it.

Honest observation. The theorems apply to the observed process itself, with its full path law, and not to models fitted to it. This distinction does real work. A first-order Markov chain fitted to an observed process, a thresholded transition graph, or a clustering of approximately stationary laws can each differ from the audited object in the very feature being claimed: the fitted chain is Markov by construction, thresholding can create or destroy cycles, and clustering under another metric can split one object into several. The theorems are silent about all three. The protocol trap shows the typical pattern: observed data can pass through recognizable stages, while the observed path law is exactly symmetric under reversal. More generally, the program of the Six Birds series [19, 21, 23] is to state which primitive, added to which finite model, produces which audited feature; results of the kind proved here mark the cases in which the answer is "none".

Scope and limitations. The setting is deliberately narrow. State spaces are finite (the graph and ladder theorems do not need this), time is discrete, and lenses are deterministic, although Remark 4.1 extends the arrow bound to memoryless noisy observation. All quantities are exact functions of a known model; estimating them from finite samples raises statistical questions that are not addressed here. The results are qualitative or give simple explicit bounds. Natural next steps include quantitative versions, such as bounds on how much arrow a given lens must lose or on how the closure deficit changes under refinement of the lens, together with continuous-time and countable-state analogues.

Formal verification. Because the theorems are meant to serve as reference points for audits, their exact hypotheses matter as much as their conclusions. For this reason all eight are formalized in Lean 4. The formal statements are in places stronger than the versions given here (Appendix E); the supporting results listed there as outside the formal development are either proved in this paper or are standard facts with references.

A Proofs for the arrow and contraction theorems

A.1 Deterministic data processing

Proof of Lemma 4.1. If $D_{\mathrm{KL}}(p \,\|\, q) = \infty$ there is nothing to prove, so assume it is finite; then $q(x) > 0$ whenever $p(x) > 0$. For $y \in \Omega'$ let $A_y = g^{-1}(y)$, $A_y^+ = \{x \in A_y : p(x) > 0\}$, $p_y = p(A_y) = p(A_y^+)$ and $q_y = q(A_y)$. The log-sum inequality [3] states that for positive numbers a_i and b_i,

$$\Big(\sum_i a_i\Big) \log \frac{\sum_i a_i}{\sum_i b_i} \le \sum_i a_i \log \frac{a_i}{b_i}.$$

Fix y with $p_y > 0$. Applying the inequality to $a_x = p(x)$ and $b_x = q(x)$ for $x \in A_y^+$, and using $q_y \ge q(A_y^+) > 0$ together with the fact that the logarithm is increasing,

$$p_y \log \frac{p_y}{q_y} \le p_y \log \frac{p_y}{q(A_y^+)} \le \sum_{x \in A_y^+} p(x) \log \frac{p(x)}{q(x)}.$$

In particular $q_y > 0$ whenever $p_y > 0$, so $D_{\mathrm{KL}}(g_{\#}p \,\|\, g_{\#}q)$ is the finite sum of the left-hand sides over y with $p_y > 0$. The sets A_y^+ partition $\{x : p(x) > 0\}$, so summing over y gives $D_{\mathrm{KL}}(g_{\#}p \,\|\, g_{\#}q) \le D_{\mathrm{KL}}(p \,\|\, q)$. □

A.2 Detailed balance and reversed paths

Proof of Lemma 4.3. Write $w(x_0,\dots,x_T) = \pi(x_0)\prod_{t=0}^{T-1} P(x_t,x_{t+1})$ and $\bar{w}(x_0,\dots,x_T) = \pi(x_T)\prod_{t=0}^{T-1} P(x_{t+1},x_t)$. We show $w = \bar{w}$ on X^{T+1} by induction on T. For $T = 0$ both equal $\pi(x_0)$. Assume the identity for paths of length T, and let $(x_0,\dots,x_{T+1})$ be a path. Then

$$\begin{aligned} w(x_0,\dots,x_{T+1}) &= w(x_0,\dots,x_T)\,P(x_T,x_{T+1}) = \bar{w}(x_0,\dots,x_T)\,P(x_T,x_{T+1}) \\ &= \Big(\prod_{t=0}^{T-1} P(x_{t+1},x_t)\Big)\,\pi(x_T)P(x_T,x_{T+1}) = \Big(\prod_{t=0}^{T-1} P(x_{t+1},x_t)\Big)\,\pi(x_{T+1})P(x_{T+1},x_T) \\ &= \bar{w}(x_0,\dots,x_{T+1}), \end{aligned}$$

where the second equality is the induction hypothesis and the fourth is detailed balance for the pair (x_T,x_{T+1}). Only multiplication is used, so states with $\pi(x) = 0$ need no separate treatment. Finally, $\mathbf{P}_T^{\pi,P}(\gamma) = w(\gamma)$ and $(\mathrm{rev}_{\#}\mathbf{P}_T^{\pi,P})(\gamma) = \mathbf{P}_T^{\pi,P}(\mathrm{rev}\gamma) = \bar{w}(\gamma)$, so the two path laws agree. □

Multiplying the T instances of detailed balance along a path instead gives

$$\pi(x_0)\cdots\pi(x_{T-1})\prod_{t=0}^{T-1} P(x_t,x_{t+1}) = \pi(x_1)\cdots\pi(x_T)\prod_{t=0}^{T-1} P(x_{t+1},x_t),$$

and cancelling the interior factors $\pi(x_1),\dots,\pi(x_{T-1})$ is valid only when they are nonzero. The induction avoids this restriction.

A.3 Total variation and Dobrushin contraction

For $\mu,\nu \in \Delta(S)$ write $\eta = \mu - \nu$ and $\eta^+ = \max(\eta,0)$ pointwise. Since $\sum_s \eta(s) = 0$, the positive and negative parts of η have equal mass, so $\sum_s \eta^+(s) = \frac{1}{2}\sum_s |\eta(s)| = d_{\mathrm{TV}}(\mu,\nu)$. For every event A, $\mu(A) - \nu(A) = \sum_{s\in A}\eta(s) \le \sum_s \eta^+(s)$, with equality for $A = \{s : \eta(s) > 0\}$. This proves the event form $d_{\mathrm{TV}}(\mu,\nu) = \max_A(\mu(A) - \nu(A))$ used below.

Proof of Lemma 7.1. Fix an event $A \subseteq S$ and choose $s_0 \in S$ that minimizes $P(s,A)$ over $s \in S$; this is possible because S is finite and nonempty. Set $c_s = P(s,A) - P(s_0,A)$. By the choice of s_0 and the event form of total variation,

$$0 \le c_s \le d_{\mathrm{TV}}\big(P(s,\cdot),P(s_0,\cdot)\big) \le \lambda(P) \qquad \text{for every } s.$$

Since $\sum_s \eta(s) = 0$, the constant $P(s_0,A)$ may be subtracted inside the sum:

$$(\mu P)(A) - (\nu P)(A) = \sum_s \eta(s)P(s,A) = \sum_s \eta(s)\,c_s.$$

Each term satisfies $\eta(s)c_s \le \lambda(P)\,\eta^+(s)$: if $\eta(s) \ge 0$ this follows from $c_s \le \lambda(P)$, and if $\eta(s) < 0$ the term is at most 0 because $c_s \ge 0$. Hence $(\mu P)(A) - (\nu P)(A) \le \lambda(P)\sum_s \eta^+(s) = \lambda(P)\,d_{\mathrm{TV}}(\mu,\nu)$. Taking the maximum over A gives the claim. □

The choice of s_0 matters. For an arbitrary reference row the coefficients c_s can be negative, and then a term with $\eta(s) < 0$ can be positive, so the terms with $\eta(s) < 0$ cannot simply be dropped.

A.4 Existence and convergence under strict contraction

Proof of Proposition 7.1. Fix μ and put $\mu_t = \mu P^t$. By Lemma 7.1 and induction, $d_{\mathrm{TV}}(\mu_{t+1}, \mu_t) \le \lambda^t d_{\mathrm{TV}}(\mu_1, \mu_0)$ for all $t \ge 0$. For $m > n$ the triangle inequality gives

$$d_{\mathrm{TV}}(\mu_m, \mu_n) \le \sum_{k=n}^{m-1} \lambda^k d_{\mathrm{TV}}(\mu_1, \mu_0) \le \frac{\lambda^n}{1-\lambda}\, d_{\mathrm{TV}}(\mu_1, \mu_0),$$

so (μ_t) is a Cauchy sequence. The simplex $\Delta(S)$ is a closed subset of $\mathbb{R}^S$ and total variation is half the ℓ^1 norm, so the sequence has a limit $\pi \in \Delta(S)$. The map $\nu \mapsto \nu P$ is linear, hence continuous, so $\pi P = \lim_t \mu_{t+1} = \pi$. Uniqueness is the case $\varepsilon = 0$ of Theorem 3.6. Finally, applying Lemma 7.1 t times with $\nu = \pi = \pi P^t$ gives $d_{\mathrm{TV}}(\mu P^t, \pi) \le \lambda^t d_{\mathrm{TV}}(\mu, \pi)$. □

B Proofs for the graph theorems

Completion of the proof of Theorem 3.3. Let ϕ be defined as in Section 5: for each vertex v, $\phi(v)$ is the sum of a along the unique path γ_v from the root of v's component to v. Let $\{u, v\}$ be an edge; u and v lie in the same component, with root r. We show $a(u, v) = \phi(v) - \phi(u)$.

Case 1: v does not lie on γ_u. Then γ_u followed by the step from u to v is a path from r to v, so by uniqueness it is γ_v. Hence $\phi(v) = \phi(u) + a(u, v)$.

Case 2: v lies on γ_u. The part of γ_u from r to v is a path from r to v, so it is γ_v. The remaining part of γ_u is a path from v to u. The single edge from v to u is also such a path, so by uniqueness of paths in a forest the remaining part is that edge, and γ_u is γ_v followed by the step from v to u. Hence $\phi(u) = \phi(v) + a(v, u) = \phi(v) - a(u, v)$ by antisymmetry.

In both cases $a(u, v) = \phi(v) - \phi(u)$. The argument uses only that each component has a root and that paths between two vertices of a forest are unique; it does not use finiteness of V. For an infinite forest, choosing one root in each component uses the axiom of choice. □

Proof of the converse of Theorem 3.4. Let G be connected, fix a root r, and suppose that the sum of the antisymmetric field a vanishes around every closed walk. Define $\phi(v)$ as the sum of a along any walk from r to v. This does not depend on the walk: if γ and γ' are two such walks, then γ followed by γ' traversed backwards is a closed walk, whose sum is the sum along γ minus the sum along γ' by antisymmetry, and this vanishes by hypothesis. For an edge $\{u, v\}$, extending a walk from r to u by the step to v gives $\phi(v) = \phi(u) + a(u, v)$. On a disconnected graph, apply the construction in each component. □

C Proofs for the closure theorem

Throughout, $I^+ = \{i : w_i > 0\}$, sums written $\sum_i$ run over I^+, and divergences take values in $[0, \infty]$ as in Section 2.

Proof of Lemma 6.1. Every p_i with $i \in I^+$ satisfies $w_i p_i \le W\bar{p}$, so $\bar{p}(\omega) > 0$ whenever $p_i(\omega) > 0$; hence each $D_{\mathrm{KL}}(p_i \,\|\, \bar{p})$ is finite, and so is the first sum on the right. Let $\Omega^+ = \{\omega : \bar{p}(\omega) > 0\}$. Note that $\bar{p}(\omega) > 0$ exactly when $p_i(\omega) > 0$ for some $i \in I^+$.

Case 1: $q(\omega) = 0$ for some $\omega \in \Omega^+$. Then $D_{\mathrm{KL}}(\bar{p} \,\|\, q) = \infty$, so the right-hand side is infinite. Choose $i \in I^+$ with $p_i(\omega) > 0$; then $D_{\mathrm{KL}}(p_i \,\|\, q) = \infty$ and the left-hand side is infinite as well.

Case 2: $q > 0$ on Ω^+. All divergences in the identity are then finite sums over Ω^+, and since every term is finite, sums may be rearranged and subtracted freely. For $i \in I^+$ and ω with $p_i(\omega) > 0$,

$$\log \frac{p_i(\omega)}{q(\omega)} = \log \frac{p_i(\omega)}{\bar{p}(\omega)} + \log \frac{\bar{p}(\omega)}{q(\omega)}.$$

Multiplying by $w_i p_i(\omega)$, summing over $i \in I^+$ and over ω with $p_i(\omega) > 0$, and using $\sum_i w_i p_i(\omega) = W\bar{p}(\omega)$ in the last term gives

$$\sum_i w_i \, D_{\mathrm{KL}}(p_i \,\|\, q) = \sum_i w_i \, D_{\mathrm{KL}}(p_i \,\|\, \bar{p}) + W \sum_{\omega \in \Omega^+} \bar{p}(\omega) \log \frac{\bar{p}(\omega)}{q(\omega)},$$

and the last sum is $D_{\mathrm{KL}}(\bar{p} \,\|\, q)$.

In both cases the identity holds. Since $W > 0$ and $D_{\mathrm{KL}}(\bar{p} \,\|\, q) \geq 0$ with equality only for $q = \bar{p}$, the left-hand side is minimized exactly at $q = \bar{p}$. □

Proof of Lemma 6.2. Group the microstates of positive mass by fiber:

$$\mathcal{L}_\tau(K) = \sum_{y:\, W_y > 0} \; \sum_{x \in f^{-1}(y),\, \mu(x) > 0} \mu(x) \, D_{\mathrm{KL}}(p_x \,\|\, K(y, \cdot)) \,.$$

A fiber with $W_y = 0$ contains no microstate of positive mass and does not appear, so its row $K(y, \cdot)$ is irrelevant. For a fiber with $W_y > 0$, apply Lemma 6.1 with weights $\mu(x)$ and laws p_x for $x \in f^{-1}(y)$, whose mixture is $\bar{p} = K^\star(y, \cdot)$. It shows that the inner sum is minimized exactly by the row $K(y, \cdot) = K^\star(y, \cdot)$, with finite minimum $\sum_x \mu(x) D_{\mathrm{KL}}(p_x \,\|\, K^\star(y, \cdot))$. Each fiber's row can be chosen independently, so $K^\star$ minimizes $\mathcal{L}_\tau$ and $\mathcal{L}_\tau(K^\star)$ is finite. □

Proof of Lemma 6.3. The joint law of (X_0, Y_0, Y_τ) gives the triple (x, y, y') mass $\mu(x)p_x(y')$ if $y = f(x)$ and mass 0 otherwise. The marginals that enter the conditional mutual information are therefore

$$\Pr(X_0 = x, Y_0 = f(x)) = \mu(x), \qquad \Pr(Y_0 = y) = W_y, \qquad \Pr(Y_0 = y, Y_\tau = y') = W_y \, K^\star(y, y')$$

for $W_y > 0$. By definition, with terms of zero joint mass omitted,

$$I(X_0; Y_\tau \mid Y_0) = \sum_{x, y'} \mu(x) p_x(y') \, \log \frac{\Pr(X_0 = x, Y_0 = f(x), Y_\tau = y') \, \Pr(Y_0 = f(x))}{\Pr(X_0 = x, Y_0 = f(x)) \, \Pr(Y_0 = f(x), Y_\tau = y')},$$

the sum running over pairs with $\mu(x)p_x(y') > 0$. For such a pair, $W_{f(x)} \geq \mu(x) > 0$ and $W_{f(x)} K^\star(f(x), y') \geq \mu(x) p_x(y') > 0$, so all four probabilities are positive and the ratio equals

$$\frac{\mu(x) p_x(y') \, W_{f(x)}}{\mu(x) \, W_{f(x)} K^\star(f(x), y')} = \frac{p_x(y')}{K^\star(f(x), y')}.$$

Summing first over y' for each x with $\mu(x) > 0$ gives $\sum_{x:\, \mu(x) > 0} \mu(x) D_{\mathrm{KL}}(p_x \,\|\, K^\star(f(x), \cdot))$, which is $\mathcal{L}_\tau(K^\star)$ by definition. □

D Proofs for the ladder theorems

Proof of Lemma 8.1. For $n = 1$ there is nothing to prove. If $e^{\circ n} = e$, then $e^{\circ(n+1)} = e^{\circ n} \circ e = e \circ e = e$. □

Proof of Lemma 8.2. Well defined and onto. For $A \subseteq \mathrm{im}(f)$, the predicate $\mathbf{1}_A \circ f$ is f-definable, with $g = \mathbf{1}_A$. Conversely, let $h = g \circ f$ be f-definable and put $A = \{y \in \mathrm{im}(f) : g(y) = 1\}$. For every x, $f(x) \in \mathrm{im}(f)$, so $h(x) = 1$ if and only if $f(x) \in A$; thus $h = \mathbf{1}_A \circ f$.

Order isomorphism. If $A \subseteq B$ then $\mathbf{1}_A \leq \mathbf{1}_B$, hence $\mathbf{1}_A \circ f \leq \mathbf{1}_B \circ f$. Conversely, suppose $\mathbf{1}_A \circ f \leq \mathbf{1}_B \circ f$ and let $y \in A$. Since $y \in \mathrm{im}(f)$ there is x with $f(x) = y$, and $1 = \mathbf{1}_A(f(x)) \leq \mathbf{1}_B(f(x))$, so $y \in B$. Thus the map reflects the order, and in particular it is injective: $\mathbf{1}_A \circ f = \mathbf{1}_B \circ f$ implies $A \subseteq B$ and $B \subseteq A$. □

E Formal verification

All eight theorems are formalized in Lean 4 (toolchain `v4.28.0`) on top of Mathlib (pinned to the matching release), in the Lean library of the accompanying repository. Table 2 lists, for each theorem, the declarations that prove it. Declaration names are relative to the library's top-level namespace, and `RP` abbreviates `RealProbability`. The library builds from a clean checkout; the listed declarations are checked against explicit target statements in a separate file; and Lean's axiom report shows that they depend only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The library contains no unproved goals, no added axioms, and no calls to native code evaluation in proofs.

Modelling choices. A finite state space is represented as `Fin n`; since every finite set is in bijection with some `Fin n`, this loses no generality. A probability law is a real-valued function that is nonnegative and sums to one. Kullback–Leibler divergence takes values in the extended reals: it is $+\infty$ exactly when the support condition fails, and otherwise the finite sum with terms of zero mass omitted. Path laws are built from the transition product, so reversal symmetry under detailed balance is proved rather than assumed. In the closure theorem the conditional mutual information is defined directly from the joint masses $\mu(x)p_x(y')$, $\mu(x)$, W_y, and $\Pr(Y_0 = y, Y_\tau = y')$, independently of the variational objective, and the candidate macro kernels range over all Markov kernels. Graphs are Mathlib's simple graphs, edge fields are real-valued, and closed-walk sums are computed along Mathlib walks.

Results outside the formal development. The following supporting results are proved, or cited as standard, in the text but are not part of the Lean library: the existence of a stationary law and the geometric convergence in Proposition 7.1, and Corollary 7.1, which uses that existence (uniqueness is formalized); Corollary 5.1; the converse of Theorem 3.4 (Appendix B); in Theorem 3.5(iii), the averaging step showing that positive-mass rows which agree pairwise within each fiber equal their fiber average, which yields $\mathrm{CD}_\tau = 0$ (the direction from pairwise disagreement to a positive deficit is formalized); the order-isomorphism part of Lemma 8.2 and the chain-length bound of Remark 8.1; the extension of the arrow bound to memoryless noisy observation in Remark 4.1; and the relation between the arrow and entropy production in Remark 4.3. Of the examples, the positive closure deficit at time 0 in Example 6.1 is checked formally.

Table 2. Formal counterparts of the eight theorems.

Theorem	Declarations	Scope of the formal statement
3.1 Arrow monotonicity	`RP.markovArrowDPI`, `RP.KL_pushforward_le`	Any initial law, kernel, lens and horizon; inequality in the extended reals.
3.2 Protocol trap	`RP.protocolTrap`, `RP.detailedBalance_pathLaw_reversible`	Detailed balance on all pairs is the only hypothesis; states of zero mass allowed.
3.3 Forest exactness	`RealGraph.forest_exact`	Any acyclic simple graph, finite or not; antisymmetry and the conclusion only on edges.
3.4 Null cycle affinity	`RealGraph.exact_closedWalk_zero`	Any simple graph; gradient condition only on edges; every closed walk.
3.5 Closure deficit	`RP.closureTheorem`, `RP.closure_variational_minimum`, `RP.conditionalInfo_positive_of_distinct_fiber_rows`; zero criterion via `RP.conditionalInfo_eq_bestObjective` and `RP.finiteObjective_eq_zero_iff`	Any current law and any lag $\tau \geq 0$. Parts (i) and (ii), with all Markov kernels as candidates, including those of infinite loss, and the positivity statement of (iii). The two declarations for the zero criterion combine to give $\mathrm{CD}_\tau = 0$ if and only if $p_x = K^\star(f(x), \cdot)$ for every x of positive mass; the step to the pairwise form stated in (iii) is not mechanized.
3.6 Contractive objecthood	`RP.dobrushin_contraction`, `RP.contractive_separation`, `RP.contractive_stationary_unique`	Nonempty finite state space; $\lambda(P)$ is the largest total variation distance between rows.
3.7 Idempotent saturation	`iterate_stabilizes_ext`	Any type.
3.8 Bounded interface	`FiniteImage.definable_cardinality`, `FiniteImage.no_infinite_distinct_sequence`	Arbitrary domain and codomain with finite image; no injective sequence $\mathbb{N} \to \mathrm{Def}(f)$.

F Version history

Version 2 (October 3, 2026).

- *Closure theorem.* In version 1 the setup and the proof section assumed a stationary chain, under which the result is correct, but the theorem statement omitted that assumption: it weighted the closure loss by an arbitrary initial law while conditioning at a later time t, which fails in general (Example 6.1). Version 2 states Theorem 3.5 for the current law at time 0, which reconciles statement and proof and removes the need for stationarity. Zero-mass microstates and fibers of zero mass are now treated explicitly, the zero criterion is stated for microstates of positive mass, and the proof of the mixture decomposition separates the case of infinite divergence.

- *Protocol trap.* Detailed balance of the initial law is the only hypothesis; stationarity follows from it. The proof that detailed balance makes path laws reversal invariant is now an induction that never divides, so it covers laws with zeros; the version 1 argument cancelled factors of π that can vanish.

- *Dobrushin contraction.* The proof of Lemma 7.1 now uses the row that minimizes the probability of the event, which makes every coefficient nonnegative; the version 1 argument used an arbitrary reference row and dropped terms whose sign it did not control. Dobrushin's coefficient is defined as the largest total variation distance between rows throughout, and the state space is assumed nonempty.

- *Scope of the other theorems.* The graph theorems are stated for edge fields defined on edges only and for graphs that need not be finite; the bounded-interface theorem is stated for arbitrary sets

with finite image and excludes every infinite sequence of distinct definable predicates. The arrow theorem allows infinite divergences and $T \geq 0$, and Remark 4.1 records that memoryless noisy observation obeys the same bound.

- *Formalization.* All eight theorems are now formalized directly in Lean, on real-valued probability laws with extended-real divergences (Appendix E). In version 1 the formal support for the objecthood theorem covered only an auxiliary uniqueness statement, and the formal support for several other theorems rested on simplified representations or took key analytic steps as hypotheses.
- *Exposition.* The text has been rewritten throughout, with a new abstract, a summary table, five figures, Corollaries 5.1 and 7.1, and further examples. The title is unchanged, and each of the eight theorems keeps its conclusion.

Version 1 (March 23, 2026). First public version.

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